



Pricing Across a Product's Lifecycle

Helping the new-to-online seller set the first price, evolve it with real signals, and keep the profit Meesho's own data says is there

DICE Challenge Season 3 • Business Track • Round 2

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Good offline sellers lose money online

1 Problem

2 Economics

3 Launch

4 Lifecycle

5 Diagnose

6 Solution

7 Engine

8 Pilot & impact

9 Limits & risks

10 ProfitPilot 2.0

1 The case question



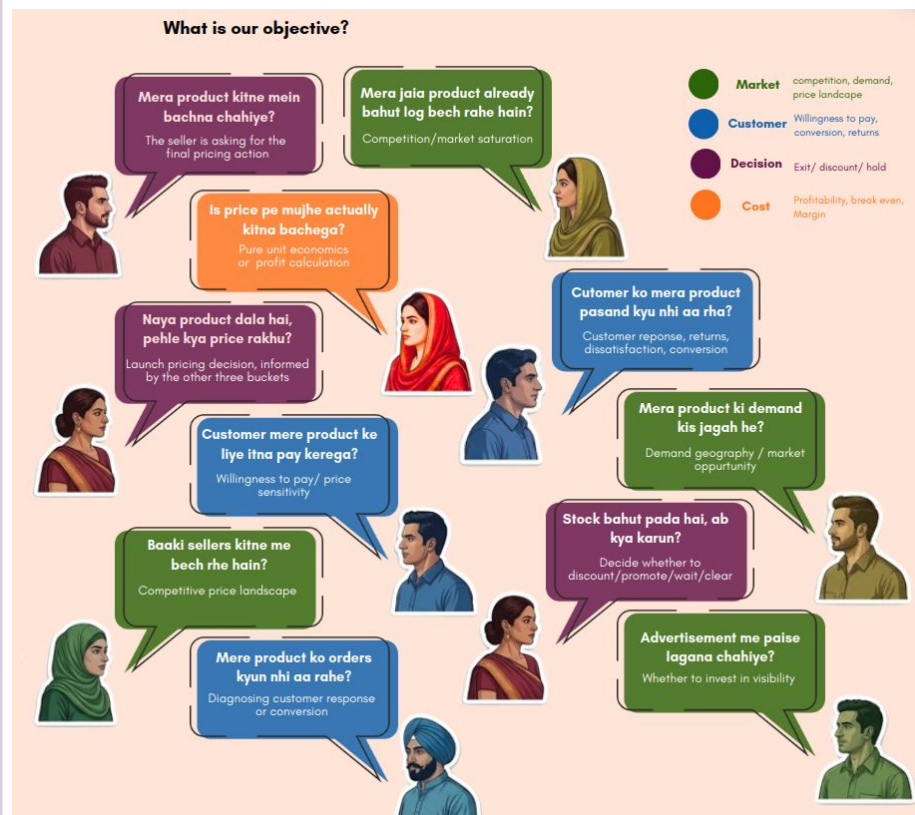
“A seller who has only ever sold offline is now listing on Meesho. How should they decide what price to set for a product, and how should that decision evolve across the product’s life so they get the most out of it? Since not every seller has the understanding, data or resources to do this well, how can Meesho help them?”

Our answer: ProfitPilot, a pricing co-pilot in the supplier panel that computes the return-adjusted floor, proposes a dual price and re-decides it at every lifecycle trigger.

2 Offline instinct vs online reality

Offline: what he knows	Online on Meesho: what changes
Price is negotiated	Buyers compare side by side; price-sensitive
Sale is final	Cancellations, RTO and returns after the sale
Customers are known	Unknown buyers; COD orders get refused
No platform costs	Freight by weight slab, packaging, ads
A few local rivals	Hundreds of near-identical listings in feed
Cash in hand	Cash locked until settlement
Haggle, bundle, bulk deal	Listed price is the whole negotiation
Word of mouth	Ads, affiliates, discounts buy rank
Fraud rare	Wardrobing, swaps, false ‘not received’

3 The seller’s questions = our objective



Every question falls in one of four buckets, each answered on a slide:

Cost → Slide 2**Market** → Slide 3**Customer** → Slide 5**Decision** → Slides 4–6

In plain English

“How much should I sell it for?”

“Will it actually make a profit?”

“Others already sell it. What now?”

“Why aren’t buyers ordering?”

“Why do buyers return it?”

“Where is the demand?”

“Stock is stuck. Discount or wait?”

“Should I pay for ads?”

4 How big the problem is

58.8%

of Meesho’s GMV survives as NMV (delivered and not returned)

[\[Choice Broking, May 2026 \(FY26\)\]](#)

7.06 lakh

annual transacting sellers, most of them small

[\[Axis Capital IPO note\]](#)

25–40%

of clothes bought online in India are returned

[\[Rest of World\]](#)

~80%

of Indian online shoppers live in Tier-2 cities and beyond (Meesho report)

[\[YourStory\]](#)

5 Why day-one pricing fails today

Carries offline price: too high online → never discovered → **floor + band check**

Copies cheapest rival: loses money on every order → **hard floor, Loss Warning**

Prices, then panics: cuts when sales dip, wrong cause → **diagnose-first tree**

Never revisits: price goes stale for months → **weekly triggers, Co-Pilot**

6 Market research: what shapes price on Meesho

Value e-commerce AOV ≈ ₹274; every rupee matters	Tier-2+ demand 80%+ of online shoppers Tier-2+	0% commission seller pays freight, not fees	COD-heavy COD orders fail 20.9% vs prepaid 5.8%
Fashion returns 25–40% of clothes come back	Valmo logistics ~50% of Meesho orders	Feed-led discovery first 2 weeks decide reach	Dual pricing return / no-return price exists

Sources: [\[Axis Capital IPO\]](#) [\[TechCrunch\]](#) [\[Rest of World\]](#) [\[FirstResort\]](#) [\[Business Standard\]](#)

7 Who we design for illustrative persona

**Ramesh, 42 · Surat**

saree & kurti trader, 18 years offline

- Prices at cost + 30%
- Counts orders, not settlement
- No time for analytics; WhatsApp-first

Needs: one number he can trust, in his language, with the reason shown.

8 Brief’s 5 challenges → our answer

Low pricing literacy → floor computed for him (2 inputs)

No history → category prior + catalogue (slide 3)

Moving target → weekly triggers & tree (slides 4–5)

Thin margins → hard floor, Loss Warning, ±8% step

Scale & diversity → category-aware models, 4 goal modes

M1 Kept-order rate $k = (1 - c)(1 - p)(1 - r)$

where c = cancel share of placed, p = RTO share of dispatched, r = return share of delivered
e.g. $0.97 \times 0.918 \times 0.876 \approx 0.78$ → only 78 of 100 placed orders ever pay the seller

M2 Ordered vs kept value $NMV = GMV \times k \times (\text{kept price} \div \text{listed price})$

where NMV counts delivered, non-returned orders only; the gap is cancels, RTO, returns and discounts
e.g. FY26 NMV/GMV ≈ 58.8% (Choice Broking) → ~41% of ordered value never settles

M3 The profit illusion $\Pi_{naive} = P - C \square$ vs $\Pi_{true} = P - F$

where P = listed price, $C \square$ = sourcing cost, F = return-adjusted floor (slide 2)
e.g. ₹399 – ₹180 = ₹219 felt vs ₹399 – ₹309 = ₹90 real → seller over-estimates profit 2.4×

KEY TAKEAWAY He works hard but prices on offline numbers; the slides that follow answer each of his questions.

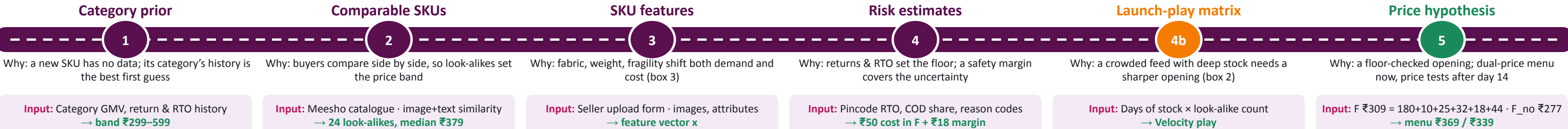
NEXT ▶ Economics

Launch: the first price is a tested hypothesis

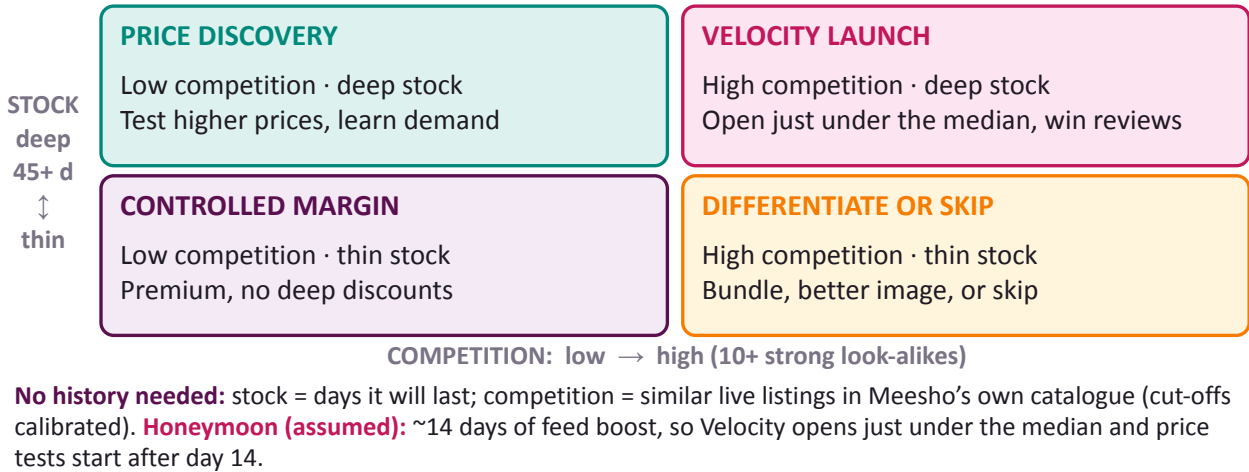


1 The cold-start path

no sales history needed; every input exists on day 0 · each step says why it is used



2 Step 4b: launch-play matrix



3 What the engine reads, by category

Category	Attributes → change the floor or price sensitivity
Apparel	fabric, print, size chart, fit
Home & kitchen	material, capacity, weight, fragility
Beauty & care	ingredients, volume, shelf life, leak risk
Kids & baby	age band, safety certification, size
Artificial jewellery	plating, stone type, weight, tarnish risk
All categories	images, price band, parcel weight slab

Artisanal detector → Controlled margin: hand-block/handloom texture, craft words (Sanganeri, Bagru, Chikankari, GI tag), craft-cluster origin, few look-alikes.

4 Worked examples + 30-day playbook

SKU (inputs on day 0)	Play	Working: floor → market → rule → price
Printed kurti, Bangalore · 24 look-alikes, deep stock	Velocity	₹309 + ₹18 margin → median ₹379 → ₹369; no-return −₹30 → ₹339
Hand-block kurti, Jaipur · 3 look-alikes, 40 units	Controlled margin	F ≈ ₹430 (assumed) → band ₹499–599 → hold ₹549, no discount
Steel lunch box, Rajkot · 6 look-alikes, 300 units	Price discovery	F ₹346 → median ₹469 → ₹449; from day 15 ₹479 on alternate days

Playbook	What happens	Gate to move on
Day 0	floor check, listing QA, compliance (BIS, MRP)	listing score ≥ 80
Days 1–7	dual-price menu ₹369 / ₹339 goes live	sanity: buyers ordering? (≈ 3)
Days 8–14	read the funnel vs similar SKUs	CTR, CVR in normal range
Days 15–30	₹369/339 vs ₹389/359, alternate days	kept rate ≥ category norm

5 Category priors the engine starts from

illustrative · derivation below

Category	Band p10–p90	Return	COD	RTO	Main risk (why it matters)
Ethnic wear	₹299–599	18%	55%	11%	size & fit → return cost
Western wear	₹249–499	16%	50%	10%	fit, colour → returns
Footwear	₹299–699	21%	60%	12%	size → highest returns
Home & kitchen	₹199–799	9%	35%	7%	damage, weight slab → freight
Beauty & care	₹149–399	7%	30%	6%	leakage, expiry → write-offs
Mobile accessories	₹99–299	6%	40%	8%	COD refusal → RTO
Artificial jewellery	₹99–399	8%	45%	9%	tarnish, look mismatch → returns

How derived: band = p10–p90 of live listings in Meesho's catalogue · RTO ≈ COD share × ~20% COD failure (Unicommerce) · fashion return + RTO 26–33% sits inside the reported 25–40%; others ≈ India's 14.9% · live: Meesho 90-day medians.

6 Day-0 catalogue read: what we use

▶ Look-alikes: how crowded → play

▶ Price spread: p25/median/p75 → band

▶ Social proof: ratings → trust gap

▶ Pressure: discounts, ad density

▶ Promise: delivery days shown

▶ Refresh: daily 14 d, then weekly

▶ Source: Meesho's own live catalogue

Printed kurti	catalogue
Look-alikes	24
p25 price	₹329
Median	₹379
p75	₹429
Avg rating	3.9★
Delivery	4–6 days

7 Fallback ladder when data is missing

Floor above market	→ don't launch at a loss: lighter pack, bundle, variant or skip
No comparable SKUs	→ category prior only · wide band
Comparables, no outcomes	→ drop them as strong evidence · widen the band
No cost entered	→ seller-provided invoice/cost proxy · add safety buffer
Implausible cost	→ flag vs category · ask once
No clicks after 1,000 views	→ fix photo & title, not the price
Clicks but no orders	→ fix page & trust: reviews, size chart, delivery

M7 Demand prior

$\ln q = \alpha_{\text{cat}} + \beta \cdot \ln p + \gamma^T x$

where q = orders/day, β = this listing's price sensitivity in a crowded feed, x = SKU features
e.g. $\beta = -3$: 10% cut → +37% orders; market implies ≈ −5.4; true β learned from pooled tests

M8 Shrink prior → SKU

$\beta' = w \cdot \beta_{\text{own}} + (1 - w) \cdot \beta_{\text{prior}}, w = n \div (n + n_0)$

where day 0: blend category + look-alikes; then move to the product's own estimate as it sells at 2+ prices
e.g. day 0: w = 0 (prior only); 600 orders over 2+ prices → w = 0.75; one price alone teaches nothing

M9 Opening price

$p_0 = \max \text{₹} \dots 9 < \text{median}, p_0 \geq F + \text{margin}$

where Velocity rule; no-return price = p_0 − return cost saved (₹10 steps); optimum $F \cdot |\beta| \div (|\beta| - 1)$ is only a cross-check
e.g. median ₹379 → ₹369 ≥ ₹309 + ₹18; no-return ₹369 − ₹30 = ₹339 (saves ₹32)

KEY TAKEAWAY On day 0 the play comes from stock and catalogue data; the opening price is floor-checked, and price tests start only after day 14.

NEXT ▶ Lifecycle

Lifecycle: re-decide the price at every trigger

1 Problem

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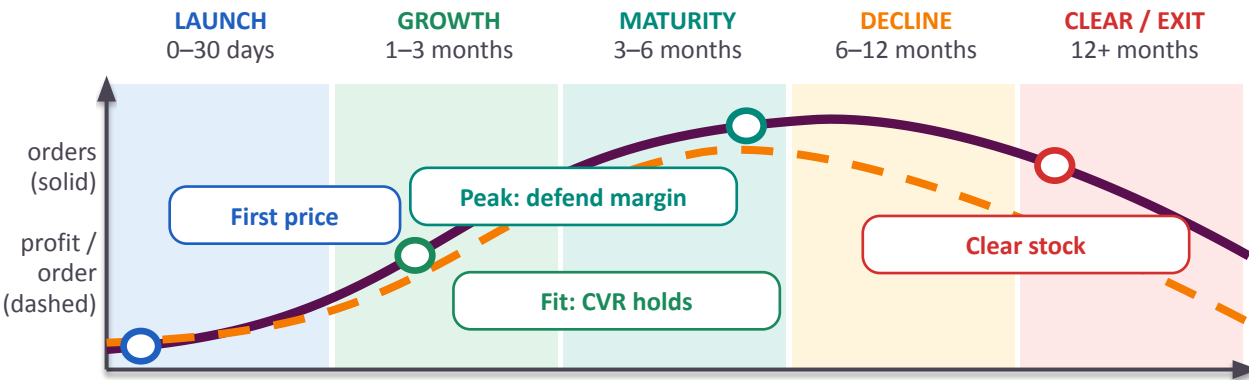
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1 The product life cycle the engine tracks orders & profit/order over time



How we drew this curve

Orders (solid) follow a Bass-like logistic S-curve: slow start, word-of-mouth take-off, saturation; then decay as copies and fatigue arrive: $q(t) = 0.04 + 0.86 \div (1 + e^{(-11(t - 0.3))}) - 2.6 \cdot \max(0, t - 0.62)^{1.7}$
Profit/order (dashed) peaks earlier and falls faster: rivals undercut, markdowns and return cost rise. Stage edges = the M10 triggers; durations are generic defaults, table 2 sets them per category. **Shape is illustrative, not fitted data.**

2 Life-cycle length differs by category planning defaults

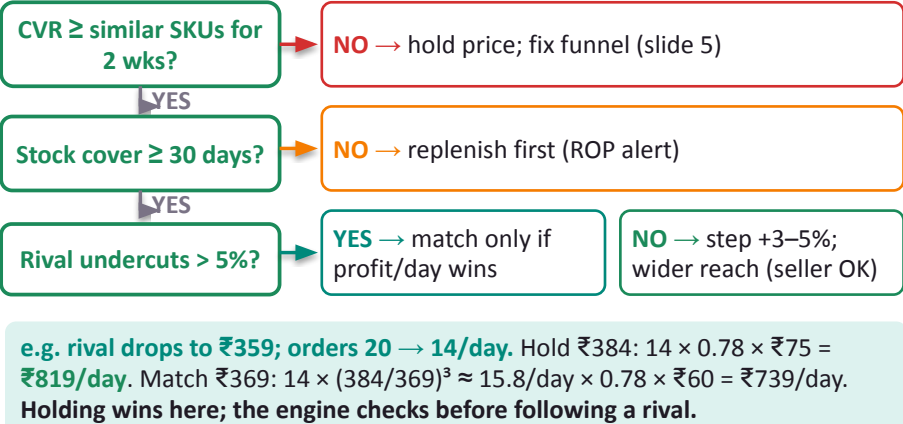
SKU	Typical life	Peak season	What ends it
Printed kurti	3–6 months	festive, wedding	design fatigue, copies
Steel lunch box	18–24 months	school reopening	demand slows; price war
Vitamin-C serum	9–12 months	year-round	shelf life / expiry
Baby romper set	6–9 months	year-round	size & season churn
Ceramic vase	12+ months	Diwali gifting	damage, overstock

So stage windows are per category: a 60-day-old kurti may already be mature, a 60-day lunch box is still launching.

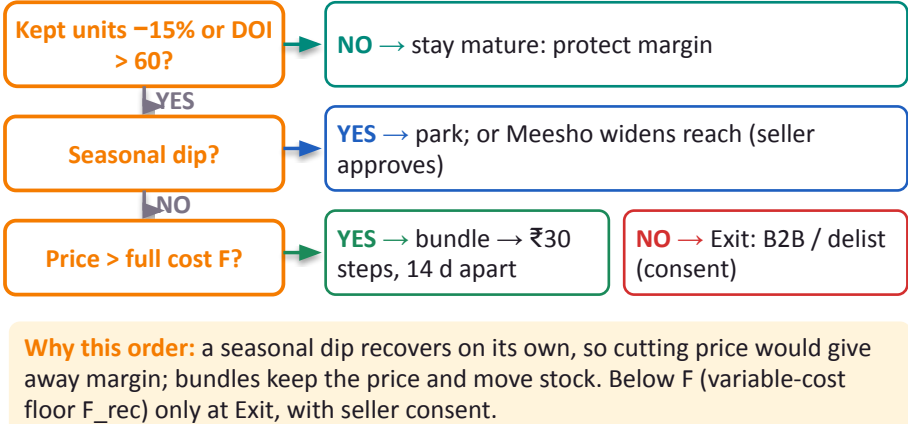
3 Signals, triggers and the price move at each stage every threshold is a default the pilot calibrates

Stage	Goal	Signals the engine reads	Trigger to act	Price move	Default mode (explained in slide 6)
Launch	Learn demand fast	Impressions, CTR, CVR, first ratings	$\geq 1,000$ impressions or 14 days	Sanity check: buyers ordering, returns normal	Co-Pilot
Growth	Scale what works	Kept-unit growth vs last season, CVR vs similar SKUs, kept rate $\geq$ category norm, rating ≥ 4.0	CVR holds 2 weeks after a step	Step up 3–5% or cut discount; Meesho widens reach to similar regions (seller approves)	Co-Pilot → Autopilot
Maturity	Protect margin	Contribution per order, impression share, rival price gap	Rival undercuts $> 5\%$ for 7 days	Defend with costs and dual price, not deep cuts	Autopilot
Decline	Recover value	Kept units -15% vs last season, days of inventory > 60 , sell-through $< 40\%$	Days of inventory > 60 or margin < 0	Bundles, wider reach (seller approves), ₹30 steps $\geq$ full cost F	Co-Pilot
Clear / exit	Free the cash	Stock age > 90 days, season end, expiry (beauty)	Season end or 90 days	Clearance, B2B liquidation; hold/relaunch next season	Seller decides

4 Growth decision tree engine · seller sees ✓/✗



5 Decline decision tree engine · seller sees ✓/✗



6 Trigger hygiene: rule + why

Max step 8% buyers notice $>10\%$ jumps; keeps the β estimate local	Min 1,000 views sanity check only: $\approx 3-5$ orders (varies by product)
7-day cooldown one full weekday + weekend cycle per test	≤ 2 moves/month no flapping; buyers & resellers trust stable prices
Never below F a sale below F loses ₹ on every kept order	Judge 14 d, confirm 28 d orders & conversion at day 14; kept orders at day 28

7 Decline exits: value recovered illustrative

Recovered = sell-through $\times$ net ₹/unit $\div$ F (vase F ₹367; one unit, % of F)

Option	Recovers	How we got it
Bundle w/ fast mover	50–80%	70–90% sold $\times$ ₹270–330 net
Markdown ladder	40–70%	60–85% sold $\times$ ₹250–300 net
Wider reach (Meesho)	40–55%	52–64% sold $\times$ (price – ₹15 hop)
B2B / bulk lot	20–50%	lot price ₹75–185 per unit
Return / donate	–11–16%	₹40–60/unit to move, no resale
Park to next season	–2%/mo	capital + space while held

M10 Stage classifier $g = U_t \div U_{\text{last season}} - 1$, DOI = stock $\div$ daily units

where U = kept units, same weeks last season (GMV moves with price) · Launch: age < 30 d · Growth: $g > +15\%$ · Maturity: $|g| \leq 15\%$ · Decline: $g < -15\%$ or DOI > 60
e.g. romper: $1,180 \div 960$ kept units $- 1 = +23\% \rightarrow$ Growth; vase DOI = $400 \div 5.5 = 73$ days $\rightarrow$ Decline

M11 Bounded step (hill-climb) $p_{(t+1)} = \text{clip}(p_{(t)}(1 \pm \eta), \pm 8\%)$ s.t. $p \geq F + \text{margin}$

where try a small step; keep it if profit/day holds at day 14 (kept orders confirm at day 28), else undo; ≤ 1 move per 7 days, 2 a month
e.g. kurti in Growth: $₹369 \times 1.04 = ₹384$ (steps are not rounded to ...9) $\rightarrow$ proposed as a Yes/No card

M12 Markdown ladder $p_k = \max(F, p_0 - k \cdot \Delta)$, $\Delta \leq 8\% \cdot p$

where Δ = rupee step ($\leq 8\%$ max step), one step per 14 days (≤ 2 moves/month); below F only at Exit, with consent, down to F_{rec}
e.g. vase: ₹499 $\rightarrow$ ₹469 $\rightarrow$ ₹439 $\rightarrow$ ₹409 $\rightarrow$ ₹379 $\rightarrow$ ₹367 (floor); Exit-only floor $F_{\text{rec}} \approx ₹260$

KEY TAKEAWAY The price is never “set and forget”: the engine moves only when data crosses a trigger, and always inside the floor.

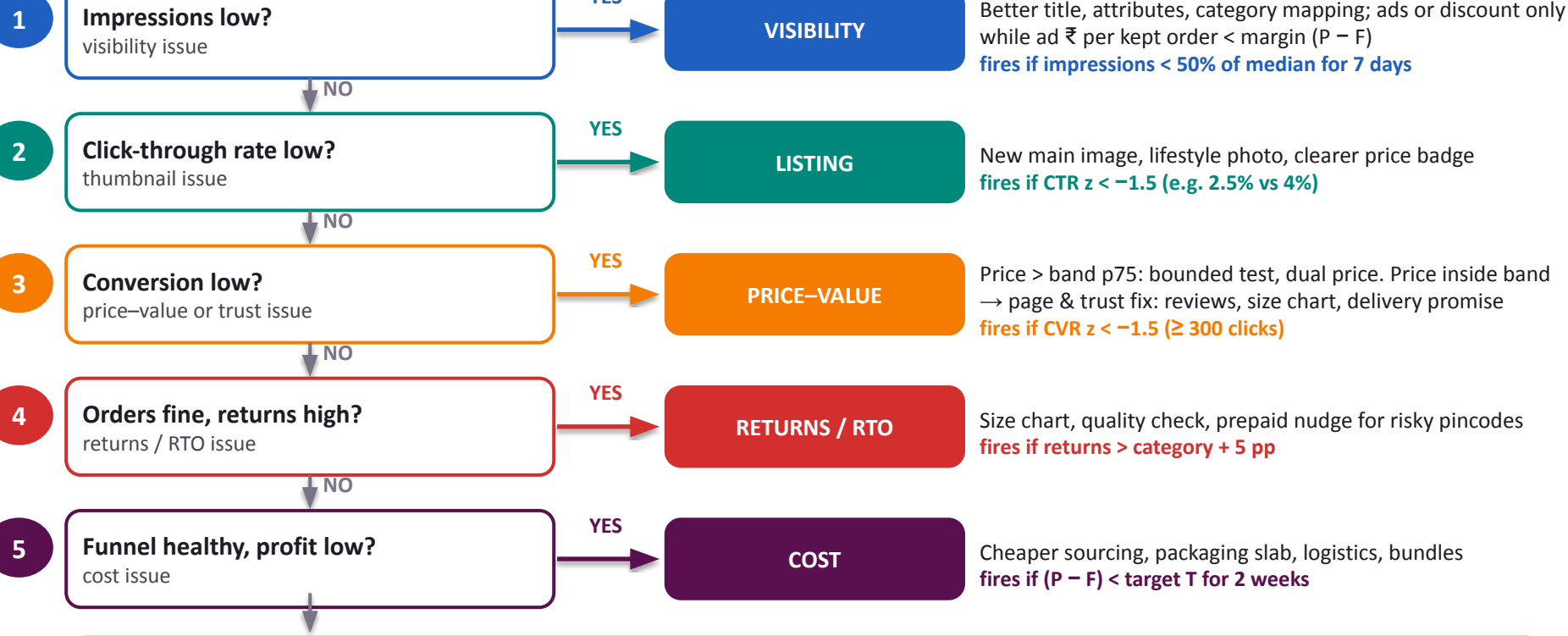
NEXT ▶ Diagnose

Diagnose before you discount

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1 The engine’s diagnostic tree

INTERNAL ENGINE LOGIC runs inside ProfitPilot · a card shows only if the signal holds 2 weeks · the seller answers ✓ Yes / ✗ No



All five NO → hold the price and scale: no discount needed

If two branches fire: fix the one with the largest ₹ loss first (M15), usually Returns > Price–value > Listing > Visibility > Cost. One fix at a time so the cause is clear.

2 What the seller sees

ProfitPilot · weekly check

Vitamin-C face serum 30 ml

This week	You	Similar
Impressions	5,000	4,000
Click rate	2.5%	3–5%
Conversion	12%	12–15%
Kept orders	86%	85–90%

People see your serum but few tap it.

The price is fine. The main photo is the problem. Try the white-background photo for 7 days?

Expected (assumed): +40% clicks ≈ 6 more orders a week, same price.

✓ Yes

✗ No

3 Panic brake for price cuts

kurti ₹369 → ₹249?

Did the price–value branch fire?
else: cut blocked

✗ no

≥ 1,000 impressions since last move?
else: wait

✓ yes

≥ 7 days since last move?
else: cooldown

✓ yes

New price ≥ return-adjusted floor?
else: loss warning

✗ ₹249

Step ≤ 8% and ≤ 2 moves a month?
else: smaller steps

✗ –33%

Profit ≥ baseline after 14 days?
else: auto-revert

—

A panic cut, priced

Kurti cut from ₹369 to ₹249 sits ₹60 below its ₹309 floor: a loss on every kept order, and the lower price pulls in more COD buyers, who refuse more.

3 of 6 checks fail → cut blocked, Loss Warning shown; the tree finds the real cause first.

Seller sees: “₹249 loses ₹60 per kept order. Keep ₹369 and let us find the real cause?”

4 Return & RTO reason codes → who fixes it

Reason code	Fix	Owner
Size / fit	measurements, size chart	Seller
Not as shown	real photos, colour check	Seller
Damaged in transit	packaging kit, handling	Meesho / Valmo
Quality / defect	QC, supplier change	Seller
Return fraud (wrong / fake item)	photo at pickup, warehouse QC	Meesho
COD refusal (RTO)	prepaid nudge, confirm call	Meesho

M13 Funnel decomposition $\Pi / \text{impression} = \text{CTR} \times \text{CVR} \times k \times (P - F)$

where CTR = clicks ÷ impressions, CVR = orders ÷ clicks, k = kept rate, P – F = margin per kept order (P, F net of GST)

e.g. serum: $0.025 \times 0.12 \times 0.86 \times (\text{₹}249 - \text{₹}166) = \text{₹}0.21$ earned per impression

M14 Which stage is broken $z_s = (x_s - \text{median}_s) \div \text{MAD}_s$; act if $z_s < -1.5$

where x_s = SKU value at stage s; median & unscaled MAD from similar SKUs; card shown only if the signal holds 2 weeks in a row

e.g. serum CTR: $(2.5\% - 4.0\%) \div 0.8\% = -1.9$, two weeks running → listing branch fires, price does not

5 Who owns RTO and returns illustrative · ethnic wear: RTO 11%, returns 18%

RTO 11% · parcel never reached the buyer → Meesho’s levers

COD refusal 8	Delivery fail 3
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Returns 18% · delivered, then sent back → mostly seller’s levers

Size 4	Image 3	Quality 3	Damage 2	Fraud 2	Other 4
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Meesho: prepaid nudge –2 pp RTO

Seller: size chart –2, photos –1 pp

6 Signal vs noise rules

- ≥ 1,000 impressions before a verdict
- CVR verdict needs ≥ 300 clicks
- Compare with similar SKUs’ median
- Signal must hold 2 weeks running
- Festival weeks flagged
- One change at a time

Serum CTR	value
SKU	2.5%
median	4.0%
MAD	0.8%
z-score	–1.9
verdict	fire (2 wks)

M15 Value of the fix $\Delta \Pi = \text{Impr} \times \Delta \text{CTR} \times \text{CVR} \times k \times (P - F)$

where the tree ranks candidate fixes by $\Delta \Pi$ and proposes only the top one to the seller

e.g. $5,000 \times 1 \text{ pp} \times 0.12 \times 0.86 \times \text{₹}83 \approx \text{₹}430$ more per week: 6 more orders (≈ 5 kept), same price

KEY TAKEAWAY Four of the five branches are not price. The tree finds the real cause; the seller just taps Yes or No.

NEXT ▶ Solution

ProfitPilot: two numbers required, a dual price out



1 In the supplier panel



Cotton printed kurti
 You enter: cost ₹180
 Target profit ₹60 / piece (pre-GST)
 rest prefilled from category defaults

EASY RETURNS
₹369
 floor ₹309

NO RETURNS
₹339
 floor ₹277

Return & RTO cost: your guess ₹15 · our estimate ₹50
₹369 is ₹42 above floor + ₹18 margin
 Expected 23 orders/month · confidence medium
 Why this price? › weekly: one Yes / No card

✓ Accept both prices

Published through Meesho's existing dual-pricing field

2 Three control modes

goals: CASH = paid sooner · GROWTH = sales first · MARGIN = ₹ per piece · CLEAR = sell fast, ≥ cost

MANUAL
Seller sets every price
Seller: types price, sees floor & band
Engine: what-if simulator, Loss Warning
Suits: day-1 sellers, 1–5 SKUs
Guard: cannot publish below floor silently
“At ₹249, after this kurti’s 12% returns (category 18%) and courier fees, you lose ₹60 per kept order.”

CO-PILOT
Default after 30 orders
Seller: taps Yes / No each week
Engine: dual price, listing & stock cards + ‘Why?’
Suits: most sellers, 5–50 SKUs
Guard: max 1 card per SKU per week
“High clicks, low conversion. Add a ₹339 no-return price: ~25% more orders (assumed) at the same ₹60.”

AUTOPILOT
For sellers with many SKUs
Seller: sets target & price range once
Engine: moves price, dual price, launch/clear
Suits: 50+ SKUs, proven trust
Guard: ±8% step · 24 h undo · 14 d check
“Target ₹60 per piece, range ₹339–₹429. Never below the ₹309 floor.”

Every suggestion has a “Why?”: what · why (the signal that fired) · ₹ effect · confidence · undo / auto-revert. e.g. “₹369 → ₹384 · CVR held 2 wks above median · +₹15 per kept order, ~12% fewer orders · confidence medium · auto-reverts if worse in 14 days”

3 Profit-protection layer: on in every mode

HARD FLOOR no mode publishes below the floor silently

PANIC BRAKE a cut needs the price–value branch to fire

LOSS WARNING every sub-floor price shows its ₹ loss first

AUTO-REVERT 14 days below baseline → back to last price

4 Trust ladder: automation is earned

After 8 weeks
Autopilot on the catalogue

After 4 accepted wins
Autopilot on 1 SKU

After 30 orders
Co-Pilot suggestions

Day 1
Manual + what-if simulator

A win = beat its holdout (alternating weeks or similar sellers).

5 Explainer video for sellers (Hindi)



Scan or open youtu.be/Mq1r7gVrDa8
Why: help offline-first sellers price without guesswork and protect margin from returns and RTO.

► **Floor & dual price:** ₹180 cost + ₹60 target → floor ₹309, price ₹369; Easy vs No Returns

► **Diagnose first:** check views, clicks, listing, size chart before any cut

► **Control:** Manual, Co-Pilot, Autopilot; hard floor, ±8% steps, 24 h undo

► **AI coach:** plain questions like “Returns kyun aa rahe hain?”

6 A seller’s first 30 days with ProfitPilot

Day 0

Day 0

Day 1

Day 7

Day 21

Day 30

Uploads kurti: cost ₹180, target ₹60
 engine: floor ₹309 + ₹18 margin

Gets dual price ₹369 / ₹339
 engine: Velocity play, dual price

Chooses Co-Pilot
 engine: weekly card scheduled

Yes to a new main photo
 engine: CTR low 2 weeks running

Accepts +4% price step
 engine: ₹369 → ₹384, ≥ floor

Early report (confirmed day 45)
 engine: +₹15/order, win #1

Seller effort: 2 inputs + 4 taps in 30 days (~5 min/week) · Outcome: profit per kept order ₹60 → ₹75, no panic cut, one Autopilot win earned

7 How the seller hears from us

Supplier panel card
 one card per SKU, Yes / No

WhatsApp digest
 weekly: ₹ kept, what changed

Local language
 Hindi and regional languages

Plain reasons
 “floor rose ₹8: returns up”



Scan: [live demo](#) (No Backend involved)

M16

Dual price $P_{\text{easy}} = F + T$, $P_{\text{no}} = (F - C_{\text{ret}}) + T$
 where T = target per piece; used if F + T sits inside the look-alike band, else show what makes T reachable (M21); the engine then fine-tunes inside the band (slide 7)
 e.g. kurti: ₹309 + ₹60 = ₹369 (inside ₹329–429); (₹309 – ₹32) + ₹60 = ₹337 → listed ₹339

M17

Loss warning $L = F - P_{\text{manual}}$ (per kept order)
 where shown before a manual price publishes; also shown as loss per month = L × kept orders
 e.g. ₹309 – ₹249 = ₹60 lost per kept order → ≈ ₹1,400 lost a month at 23 kept orders/month

M18

Trust-ladder rule promote if wins ≥ 4 and lift > 0 (90% conf.)
 where a win = accepted suggestion that beat its holdout (alternating weeks, or similar sellers) on profit per impression; 2 bad weeks in a row demote
 e.g. Co-Pilot → Autopilot on 1 SKU after 4 wins; back to Co-Pilot after 2 bad weeks ($\Pi < -10\%$)

The engine: from Meesho’s data to a 1-tap answer

1 Problem

2 Economics

3 Launch

4 Lifecycle

5 Diagnose

6 Solution

7 Engine

8 Pilot & impact

9 Limits & risks

10 ProfitPilot 2.0

1 Five layers behind every recommendation all inputs already exist inside Meesho

DATA · what Meesho already holds

- Orders, cancels, RTO, returns + reason codes
- Impressions, clicks, CVR; inventory, settlement
- Pincode & COD history; live catalogue

trace: this kurti: 12% returns, 8% RTO (cat. 18%, 11%)

FEATURES · computed per SKU-week

- Return-adjusted floor F & buffer B
- Funnel rates vs category median
- Look-alikes, artisan score, days of inventory

trace: F ₹309 + ₹18 margin = ₹327 min

MODELS · predict each branch

- Demand & elasticity, return / RTO risk
- Look-alike embeddings, stage, anomaly
- Full list in box 2 (1.0 + 2.0)

trace: $\beta = -3$, P(return) = 12%

DECISION · search inside guardrails

- Bandit over menu arms, rotated by day
- Diagnose-first tree (slide 5)
- 8 pre-flight checks (box 3)

trace: menu ₹369/339 chosen, ₹299 blocked

ACTION · what the seller sees

- Dual-price card + “Why?”
- One Yes / No suggestion a week
- Mode-aware Co-Pilot / Autopilot moves

trace: card: ₹369 / ₹339 → ✓ Yes

2 Model stack: the intelligence each decision needs pilot needs 4: floor, return risk, look-alikes, price search; the rest are rules · 2.0 = opt-in (slide 10)

Model	Decision it powers	Method	Key inputs	Refresh	Target	Ver
Cost & floor model	floor F, buffer B per SKU	cost allocation per kept order (M4, M6)	rate cards, reason codes	daily	±5% vs settlement	1.0
Demand & elasticity	orders at each price (β)	LightGBM (monotone in price) + hierarchical Bayes	price, rank, season, look-alikes	daily	MAPE $\leq$ 25%	1.0
Return / RTO risk	buffer per SKU $\times$ pincode	gradient-boosted classifier	category, size, pincode, COD	daily	AUC $\geq$ 0.75	1.0
Look-alike & artisan	competition, craft score, band	CLIP image + text embeddings, kNN	images, title, attributes	weekly	precision $\geq$ 85%	1.0
Lifecycle stage	stage $\rightarrow$ price move	trigger rules (M10) + HMM smoothing	kept-unit trend, DOI, age	daily	$\geq$ 85% vs review	1.0
Funnel anomaly	which stage is broken	robust z vs similar SKUs, 2 weeks running	CTR, CVR, kept rate	weekly	false fire $\leq$ 10%	1.0
Price bandit	which menu arm today	Thompson sampling, arms rotate by day (M20)	profit per impression	per batch	0 sub-floor moves	1.0
Fix ranker	one card a week, top ₹ first	expected $\Delta \Pi$ ranking (M15) $\times$ confidence	outputs of models above	weekly	accepted $\geq$ 20%	1.0
“Why?” explainer	plain reason in Hindi / English	templates + grounded LLM, no free numbers	fired signal, ₹ effect	per card	0 ungrounded ₹	1.0
Lift measurement	did the suggestion work?	holdout difference + CUPED	pre-period profit	weekly	detects ₹10 lift	1.0
Regional demand	pincodes, hub stock	LightGBM / Prophet per region	pincode history, festivals	weekly	MAPE $\leq$ 20%	2.0
COD risk score	prepaid nudge / confirm call	logistic regression + GBM	pincode, buyer history, value	per order	AUC $\geq$ 0.75	2.0
Fraud & abuse	wardrobing, swaps, false claims	isolation forest + buyer–seller graph	return patterns, devices	daily	precision $\geq$ 90%	2.0
Weight / size mismatch	slab re-charge, kit nudge	DWS scan vs declared, ± 50 g / ± 1 cm	hub scans (< 1 s)	per parcel	false flag $\leq$ 2%	2.0
Packaging-kit recommender	right-size, fragile kit	kNN on dimensions + damage history	dims, fragility, claims	weekly	damage 6% $\rightarrow$ 3%	2.0
Pooled procurement	group buys that clear MOQ	mixed-integer program (MILP)	forecasts, MOQ, quotes	weekly	MOQ met	2.0
Credit risk (PD)	working-capital limit	GBM scorecard, NBFC-validated	settlements, returns, turns	monthly	Gini $\geq$ 0.5	2.0
Return grading	resell / refurbish / B2B lot	CNN on hub photos	returned-unit images	per unit	accuracy $\geq$ 90%	2.0

3 Eight pre-flight checks on every move a move publishes only if all pass

Floor
price $\geq$ F

Volume
 $\geq$ seller minimum

Sample
 $\geq$ 1,000 views

Range
 $\leq$ 15% outside seen

Cost sanity
vs category range

Volatility
 $\leq$ 2 moves / month

Dispersion
no price herding

Fairness
same price, all buyers

M19 Reward $r = \text{kept} \times (P - F) \div \text{impressions}$

where profit per impression after returns & RTO; returns are credited back when the return window closes

e.g. arm ₹369: 0.0021 kept/impression $\times$ ₹60 = ₹0.126 per impression

M20 Constrained Thompson sampling $a^* = \text{argmax}_{\{P_a \geq F + m\}} \theta_a(P_a - F)$, $\theta_a \sim \text{Beta}(a, b)$

where θ_a = kept-conversion of menu arm a; starting beliefs from the pooled demand model; each day a += kept, b += misses

e.g. arms rotate by day, so all buyers see one price at a time; ₹299 never tried

M21 Lever search $\min \sum \text{cost}_j \text{ s.t. } F - \sum \Delta F_j \leq P - T$

where when no price clears target T, find the cheapest cost levers j that do; shown as 2.0 suggestions (slide 10), seller decides

e.g. kurti: F = ₹309 $\rightarrow$ ₹261 (–₹48) $\rightarrow$ ₹60 target reachable at ₹321 instead of ₹369

4 Price search that cannot lose money

Menu (easy / NR)	Role	Status
₹299 / ₹269	—	below floors ₹309 / ₹277: never tried
₹369 / ₹339	lean	eligible · most days by day 30
₹399 / ₹369	balanced	eligible
₹429 / ₹399	margin	eligible

Arms rotate by day: at any moment every buyer sees the same menu; each arm keeps the computed ₹30 gap. **Starting beliefs come from the pooled demand model.** Reward = profit per impression (M19); 5% holdout.

5 Which programme fits this seller? routes to slide 10

ENGINE LOGIC seller only sees the result as a ✓ Yes / ✗ No card

Sales history < 30 days?

YES $\rightarrow$ STARTER PACK

NO

Artisan score $\geq$ 0.7?

few copies $\rightarrow$ MAKER

many $\rightarrow$ HUMAN REVIEW

NO

Days of inventory > 60?

YES $\rightarrow$ STOCK RECOVERY

NO $\rightarrow$ FAST MOVERS

6 Running it safely (MLOps)

Rollout: shadow mode $\rightarrow$ 5% canary $\rightarrow$ all · **Drift:** PSI > 0.2 $\rightarrow$ retrain · **Fail-safe:** model down $\rightarrow$ category prior + floor · **Audit:** model cards, decision log · **Stack:** feature store, batch + Kafka, model API, A/B

KEY TAKEAWAY Each suggestion shows its reason, ₹ effect and confidence, never goes below the floor, and uses only data Meesho already has.

NEXT ▶ Pilot & impact



1 Four programmes: they attach to products, so a seller can be in more than one (slide 7 tree) pricing items already run in 1.0; the rest are 2.0

STARTER PACK

New, often offline sellers

- Onboarding + catalogue template
- Category packaging kit
- Dual-price guidance at upload
- Starting price + margin-floor guidance

Engine trigger: No sales history yet
KPI: first 10 kept orders in 30 days, vs holdout

e.g. Ramesh’s 1st kurti kit + template → live in 2 days

fixes limits 3, 7

FAST MOVERS

Deep stock, many look-alikes

- Replenishment alerts (reorder point)
- Pooled procurement (suggestion)
- Meesho widens reach (seller approves)
- Test +3–5% price steps when demand remains strong

Engine trigger: Look-alikes high, demand stable
KPI: Stock-outs –20 to –30%

e.g. Storage container set

ROP 102 → ‘Reorder ~170?’ (12/day × 14 d) ✓

fixes limit 4

MAKER PROGRAMME

Handmade, regional, small batch

- “Handcrafted” badge & premium slot, after craft / GI check
- Storytelling templates
- No discount-led positioning

Engine trigger: Artisan score ≥ 0.7, few look-alikes
KPI: Higher price realised vs commodity

e.g. Jaipur hand-block kurti

₹549 held, badge on

fixes limit 8

STOCK RECOVERY

Ageing or slow-moving stock

- Bundles with fast movers
- Regional redistribution
- Markdown steps ≥ full cost F

Engine trigger: Days of inventory > 60
KPI: Value recovered vs no action

e.g. Ceramic vase, DOI 73 d

₹499 → ₹469 → ₹439

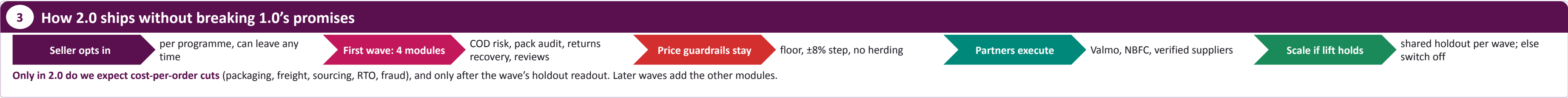
steps

fixes limits 4, 7

2 2.0 solutions: who does the work, how it runs, what it is worth, which limit it fixes AI = intelligence layer · AUTO = platform automation · OPS = Meesho team · PARTNER = regulated partner

Solution	Example SKU	Who	How it runs	Impact (basis) Illustrative	Fixes limit
Packaging + weight audit	Ceramic vase ₹499	AUTO + AI	Scan station at Valmo hub checks size & weight vs declared; mismatch → slab re-charge + kit nudge	Damage 6% → 3% with fragile kit (kit trial)	4, 7
Pooled procurement	Steel lunch box ₹449	AI + OPS	Engine pools 5 sellers’ forecast to meet a 300-unit MOQ; ops signs only once commitments are firm	₹210 → ₹188 bulk quote; floor ₹346 → ₹324	4, 7
Optional Valmo fulfilment pilot	Baby romper set ₹349	AI + AUTO	Fast movers only, opt-in: regional forecast pre-positions stock with Valmo partners; routing picks the nearest	Delivery 5.2 → 3.1 days; RTO 8% → 5% (fewer refusals)	1, 7
COD risk score	All SKUs	AI + AUTO	High-risk pincode orders get a prepaid offer or confirmation call before dispatch	COD orders fail 20.9% vs prepaid 5.8% [Unicommerce]	7
Inventory credit	Vitamin-C serum ₹249	PARTNER	Credit line sized from sales & return risk (M30); repaid through the NBFC’s own collection (lender auto-debit)	₹34k line → ~500 units; no stock-outs at peak	3, 4
Fraud & abuse shield	Kurti (wardrobing)	AI + AUTO	Abuse score; photo-verified pickup; OTP proof; price same for all, only return checks differ	Protects ~1–2 pp of returns (assumption)	7
Returns recovery	All returned units	AUTO + OPS	Grade A/B/C at hub → resell, refurbish or B2B lot; restock to nearest region	Recovers 50–80% of returned value	7
Product–customer fit	Kurti sizes	AI	Show the SKU first to buyers with high keep-probability (size history, past returns)	Fewer size returns; target –20% (assumption)	1, 8
Review intelligence	Romper reviews	AI	Indic-language topic model on reviews → listing or product fix card (“runs small”)	Catches a defect weeks before returns peak	5, 8
Root-cause engine	Any falling SKU	AI	Causal uplift on past actions (M26) names the one action that actually worked	Advice based on cause, not correlation	2
AI seller coach	All sellers	AI	Hindi / regional chat: “returns are 22%; do this next”; tracks whether advice was applied	Closes the loop: applied vs ignored	5
Lifetime-value objective	All sellers	AI	Optimise profit × survival (M25), rather than only this month’s orders	Fewer sellers quit after one bad month	6
Seller capacity profile	All sellers	AI	Capital, stock and risk appetite set the default goal mode and order sizes	Advice fits what the seller can execute	3

Sources: [\[vMeasure: scan stations <1 s/parcel\]](#) [\[FirstResort citing Unicommerce: COD vs prepaid\]](#) [\[Meesho Finance: NBFC partners\]](#) [\[Business Standard: Valmo ~50% of orders\]](#)



M28 Reorder point $ROP = \bar{d} \cdot L + z \cdot \sigma_d \cdot \sqrt{L}$

where $\bar{d}$ = avg daily demand, L = lead time (days), σ_d = daily demand std-dev, z = 1.65 for 95% service

e.g. storage set: $12 \times 7 + 1.65 \times 4.1 \times \sqrt{7} = 84 + 18 = 102 \rightarrow \text{‘Reorder ~170?’ (12/day} \times 14 \text{ days)}$

M29 Pooled procurement $c(Q) = c_{\text{bulk}}$ if $\sum_i q_i \geq \text{MOQ}$, else c_{retail}

where the engine pools forecast demand q_i of sellers buying the same item to clear the factory MOQ

e.g. 5 sellers $\times 80/70/60/50/40 = 300 \geq \text{MOQ } 300 \rightarrow \text{₹210} \rightarrow \text{₹188 per unit for all}$

M30 Credit limit $\text{Limit} = \min(2 \cdot S, V_{\text{next}}) \times (1 - r)$

where S = avg monthly settlement, V_{next} = next stock order value, r = return-rate haircut; NBFC lends and collects

e.g. serum: $\min(2 \times \text{₹25k}, \text{₹35k}) \times 0.96 \approx \text{₹34k} \rightarrow \sim 500 \text{ units}$

KEY TAKEAWAY 1.0 fixes the price; 2.0 lowers the costs behind it, with seller opt-in, partners running it, rollout in waves and a holdout to measure it.